THE COMPOSITION OF A GROUPOID OVER A FIELD IS AN OPEN MORPHISM

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The pull diagram relating the composition with the projections: commutative diagram in which the two top rows are isomorphic via the displayed vertical maps (more-groupoids chapter, equation-pull, carried as context span 301-320). cited at proof lines 1162-1164. Identifies the composition c with the projection pr_1 : R x_{t,U,t} R to R up to isomorphism.
- A morphism with target the spectrum of a field is universally open (Morphisms chapter, Lemma morphisms-lemma-scheme-over-field-universally-open, canonical tag 0383). cited at proof lines 1165-1166. Gives that the projection pr_1 : R x_{t,U,t} R to R, whose target U = Spec(k), is open.

Source reference. Original lemma 0383: sources/stacks-chow/tags.txt, source lines 4222–4222. Cited at proof lines 1165-1166: the projection pr_1 : R x_{t,U,t} R to R, with target the spectrum of a field, is open.

Lemma 0.2. *Let S be a scheme. Let (U, R, s, t, c) be a groupoid scheme over S. If U is the spectrum of a field, then the composition morphism $c : R \times_{s,U,t} R \to R$ is open.*

$$
\begin{array}{ccccc}
R \times_{t,U,t} R & \xrightarrow[\mathrm{pr}_0]{\mathrm{pr}_1} & R & \xrightarrow{t} & U \\
\downarrow{\scriptstyle \mathrm{pr}_0 \times c \circ (i,1)} & & \downarrow{\scriptstyle \mathrm{id}_R} & & \downarrow{\scriptstyle \mathrm{id}_U} \\
R \times_{s,U,t} R & \xrightarrow[\mathrm{pr}_0]{c} & R & \xrightarrow{t} & U \\
\downarrow{\scriptstyle \mathrm{pr}_1} & & \downarrow{\scriptstyle s} & & \\
R & \xrightarrow[t]{s} & U & &
\end{array}
\tag{0.2.1}
$$

is commutative. The two top rows are isomorphic via the vertical maps given. The two lower left squares are cartesian.

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Proof. The composition is isomorphic to the projection map $\mathrm{pr}_1 : R \times_{t,U,t} R \to R$ by Diagram (0.2.1). The projection is open by Morphisms, Lemma 0383. $\square$